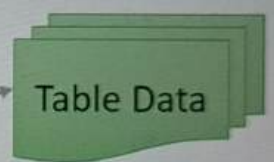
Spark Database and Tables

Database

I. Tables

II. Views



Spark Warehouse

Catalog Metastore

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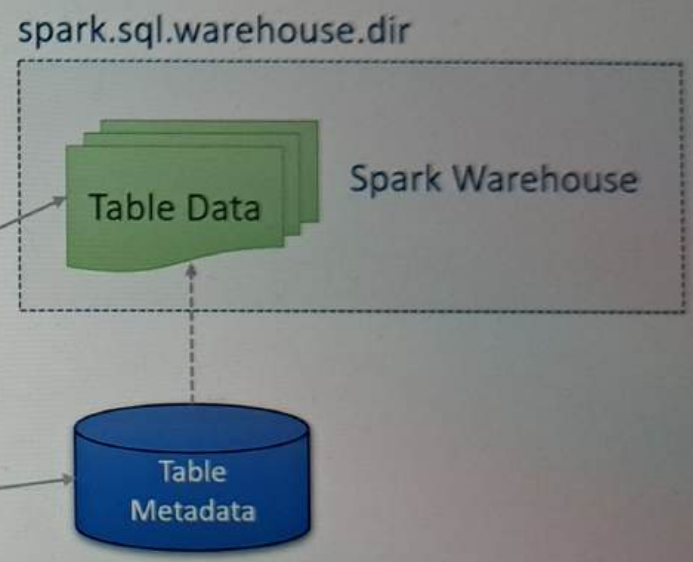
Spark Database and Tables

How to Create Managed Table?

```
dataframe.write
        .saveAsTable("your_table_name")
```



Spark Tables

I. Managed Tables
II. Unmanaged Tables (External Tables)

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Spark Database and Tables

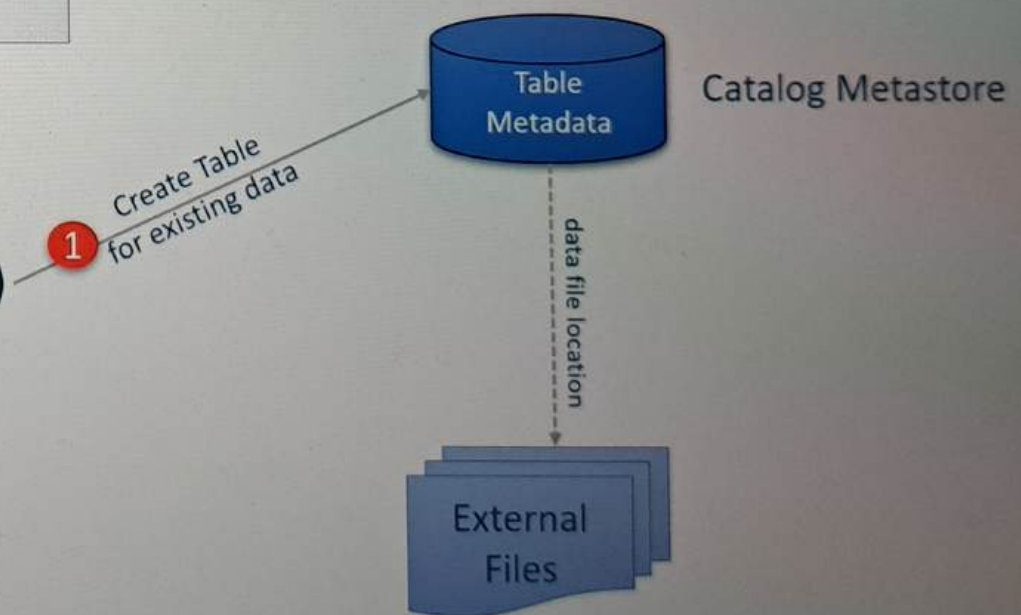
How to Create Unmanaged Table?

```
CREATE TABLE your_tbl_name (col1  data_type,
                            col2  data_type,
                            ......)
USING PARQUET
LOCATION "data_file_location"
```

Spark Tables

I. Managed Tables

II. Unmanaged Tables (External Tables)



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Spark Database and Tables

SELECT count(*) FROM ...

Spark Cluster

Spark SQL Engine

Table Metadata

External Files

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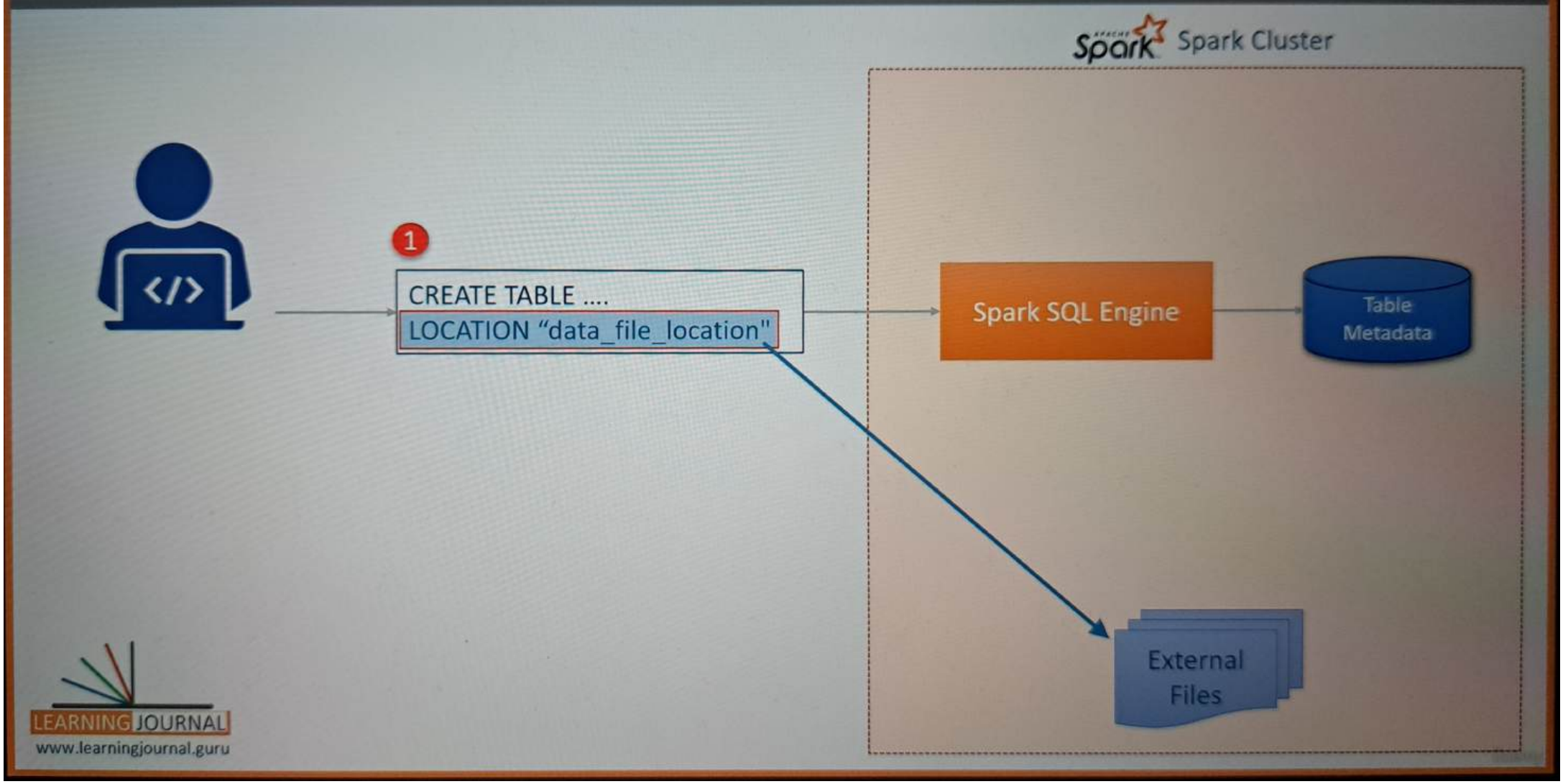
Spark Database and Tables



Spark Database and Tables

2

SELECT count(*) FROM ...

Spark Cluster

Spark SQL Engine

Table Metadata

External Files

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File Edit View Navigate Code Refactor Run Tools VCS Window Help 06-SparkSQLTableDemo - SparkSQLTableDemo.py - PyCharm

06-SparkSQLTableDemo 〉 SparkSQLTableDemo.py

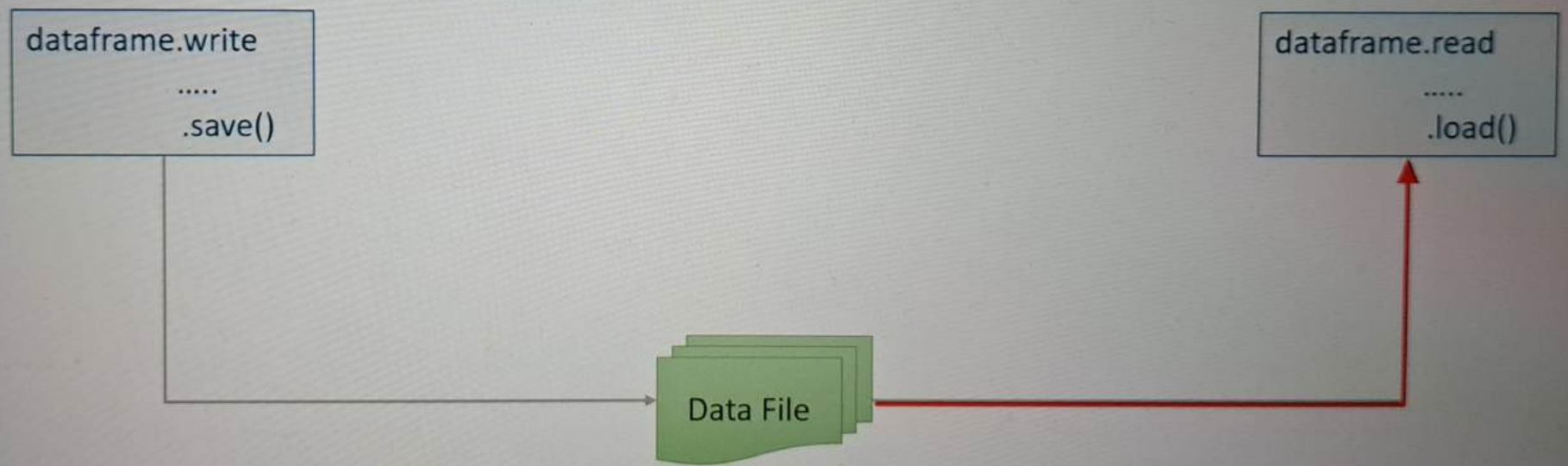
Project

- 06-SparkSQLTableDemo C:\Users\prashant\Do
  - dataSource
  - lib
  - log4j.properties
  - SparkSQLTableDemo.py
- External Libraries
- Scratches and Consoles

SparkSQLTableDemo.py

```python
from pyspark.sql import *

from lib.logger import Log4j

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .master("local[3]") \
        .appName("SparkSQLTableDemo") \
        .enableHiveSupport() \
        .getOrCreate()

    logger = Log4j(spark)

    flightTimeParquetDF = spark.read \
        .format("parquet") \
        .load("dataSource/")
```

Spark Database and Tables



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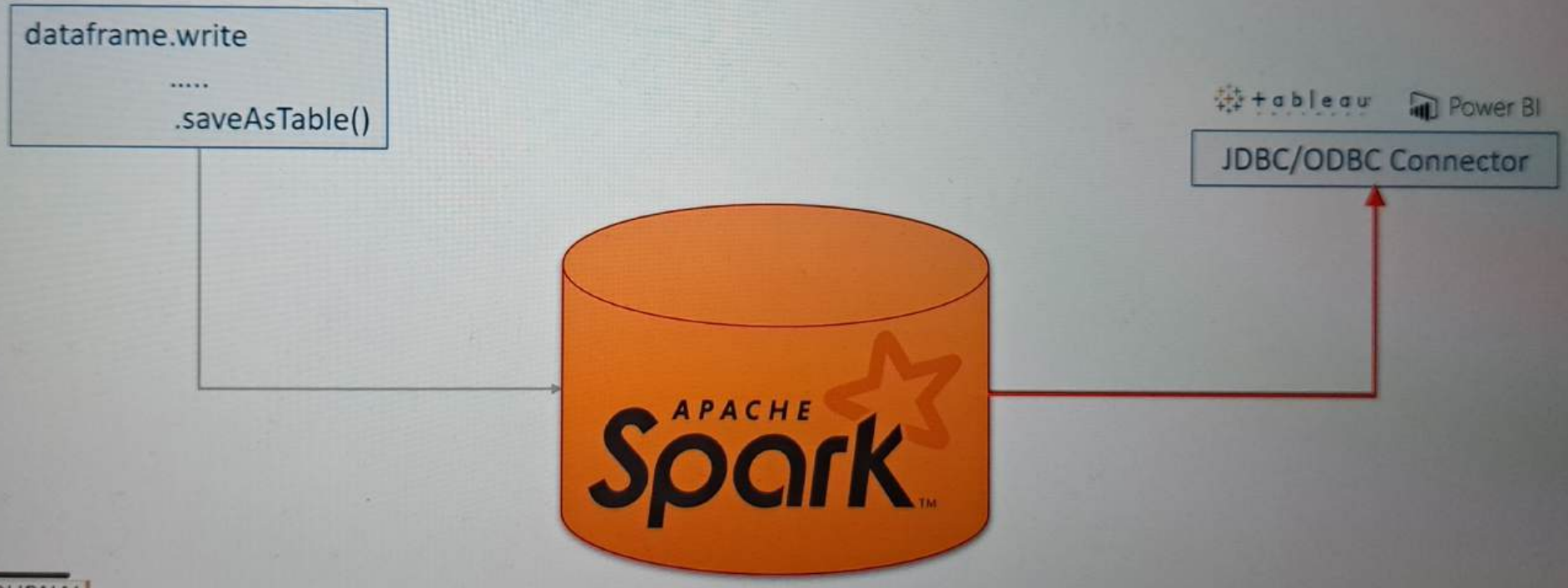
Spark Database and Tables



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```
        .builder \
        .master("local[3]") \
        .appName("SparkSQLTableDemo") \
        .enableHiveSupport() \
        .getOrCreate()

    logger = Log4j(spark)

    flightTimeParquetDF = spark.read \
        .format("parquet") \
        .load("dataSource/")

    spark.sql("CREATE DATABASE IF NOT EXISTS AIRLINE_DB")
    spark.catalog.setCurrentDatabase("AIRLINE_DB")

    flightTimeParquetDF.write \
        .mode("overwrite") \
        .saveAsTable("flight_data_tbl")

    logger.info(spark.catalog.listTables("AIRLINE_DB"))
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 06-SparkSQLTableDemo - SparkSQLTableDemo.py - PyCharm

06-SparkSQLTableDemo 〉 SparkSQLTableDemo.py

Project

- 06-SparkSQLTableDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Python\06
  - app-logs
  - dataSource
  - lib
  - metastore_db
  - spark-warehouse
    - airline_db.db
      - flight_data_tbl
        - ._SUCCESS.crc
        - .part-00000-126c21b7-f286-472f-9000-8a6dca010519-c000.snappy.parquet.crc
        - _SUCCESS
        - part-00000-126c21b7-f286-472f-9000-8a6dca010519-c000.snappy.parquet
  - derby.log
  - log4j.properties
  - SparkSQLTableDemo.py
- External Libraries
- Scratches and Consoles

SparkSQLTableDemo.py

```
10          .enableHiveSupport() \
11          .getOrCreate()
12
13      logger = Log4j(spark)
14
15      flightTimeParquetDF = spark.r
16          .format("parquet") \
17          .load("dataSource/")
18
19      spark.sql("CREATE DATABASE IF
20      spark.catalog.setCurrentDatab
21
22      flightTimeParquetDF.write \
23          .mode("overwrite") \
24          .saveAsTable("flight_data
25
26      logger.info(spark.catalog.lis
27
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help 06-SparkSQLTableDemo - SparkSQLTableDemo.py - PyCharm

06-SparkSQLTableDemo > SparkSQLTableDemo.py

Project

- 06-SparkSQLTableDemo C:\Users\prashant\Documents\(
  - app-logs
  - dataSource
  - lib
  - metastore_db
  - spark-warehouse
    - airline_db.db
      - flight_data_tbl
        - ORIGIN=ABE
        - ORIGIN=ABQ
        - ORIGIN=ADQ
        - ORIGIN=AGS
        - ORIGIN=AKN
        - ORIGIN=ALB
        - ORIGIN=AMA
        - ORIGIN=ANC
        - ORIGIN=ATL
        - ORIGIN=AUS
        - ORIGIN=AVL
        - ORIGIN=AVP
        - ORIGIN=AZO
        - ORIGIN=BDL
        - ORIGIN=BET
        - ORIGIN=BGM
        - ORIGIN=BHM
        - ORIGIN=BIL
        - ORIGIN=BIS
        - ORIGIN=BNA
        - ORIGIN=BOI
        - ORIGIN=BOS
        - ORIGIN=BRW
        - ORIGIN=BTR
        - ORIGIN=BTV
        - ORIGIN=BUF
        - ORIGIN=BUR

SparkSQLTableDemo.py

```python

from lib.logger import Log4j

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .master("local[3]") \
        .appName("SparkSQLTableDemo") \
        .enableHiveSupport() \
        .getOrCreate()

    logger = Log4j(spark)

    flightTimeParquetDF = spark.read \
        .format("parquet") \
        .load("dataSource/")

    spark.sql("CREATE DATABASE IF NOT EXISTS AIRLINE_DB")
    spark.catalog.setCurrentDatabase("AIRLINE_DB")

    flightTimeParquetDF.write \
        .mode("overwrite") \
        .partitionBy("ORIGIN", "OP_CARRIER") \
        .saveAsTable("flight_data_tbl")

    logger.info(spark.catalog.listTables("AIRLINE_DB"))
```

if __name__ == "__main__"

Git TODO Run Terminal Python Console Event Log

24:47 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

```python
flightTimeParquetDF = spark.read \
    .format("parquet") \
    .load("dataSource/")

spark.sql("CREATE DATABASE IF NOT EXISTS AIRLINE_DB")
spark.catalog.setCurrentDatabase("AIRLINE_DB")

flightTimeParquetDF.write \
    .format("csv") \
    .mode("overwrite") \
    .bucketBy(5, "OP_CARRIER", "ORIGIN") \
    .saveAsTable("flight_data_tbl")

logger.info(spark.catalog.listTables("AIRLINE_DB"))
```

06-SparkSQLTableDemo / SparkSQLTableDemo.py

Project

- metastore_db
- spark-warehouse
  - airline_db.db
    - flight_data_tbl
      - _SUCCESS.crc
      - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00000.c000.csv.crc
      - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00001.c000.csv.crc
      - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00002.c000.csv.crc
      - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00003.c000.csv.crc
      - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00004.c000.csv.crc
      - _SUCCESS
      - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00000.c000.csv
      - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00001.c000.csv
      - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00002.c000.csv
      - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00003.c000.csv
      - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00004.c000.csv
  - derby.log
  - log4j.properties
  - SparkSQLTableDemo.py
- External Libraries
- Scratches and Consoles

SparkSQLTableDemo.py

```
16          .format("parquet") \
17          .load("dataSource/")
18
19      spark.sql("CREATE DATABASE IF NOT EXISTS AIRLINE_DB")
20      spark.catalog.setCurrentDatabase("AIRLINE_DB")
21
22      flightTimeParquetDF.write \
23          .format("csv") \
24          .mode("overwrite") \
25          .bucketBy(5, "OP_CARRIER", "ORIGIN") \
26          .saveAsTable("flight_data_tbl")
27
28      logger.info(spark.catalog.listTables("AIRLINE_DB"))
29
```

if __name__ == "__main__"

Run: SparkSQLTableDemo

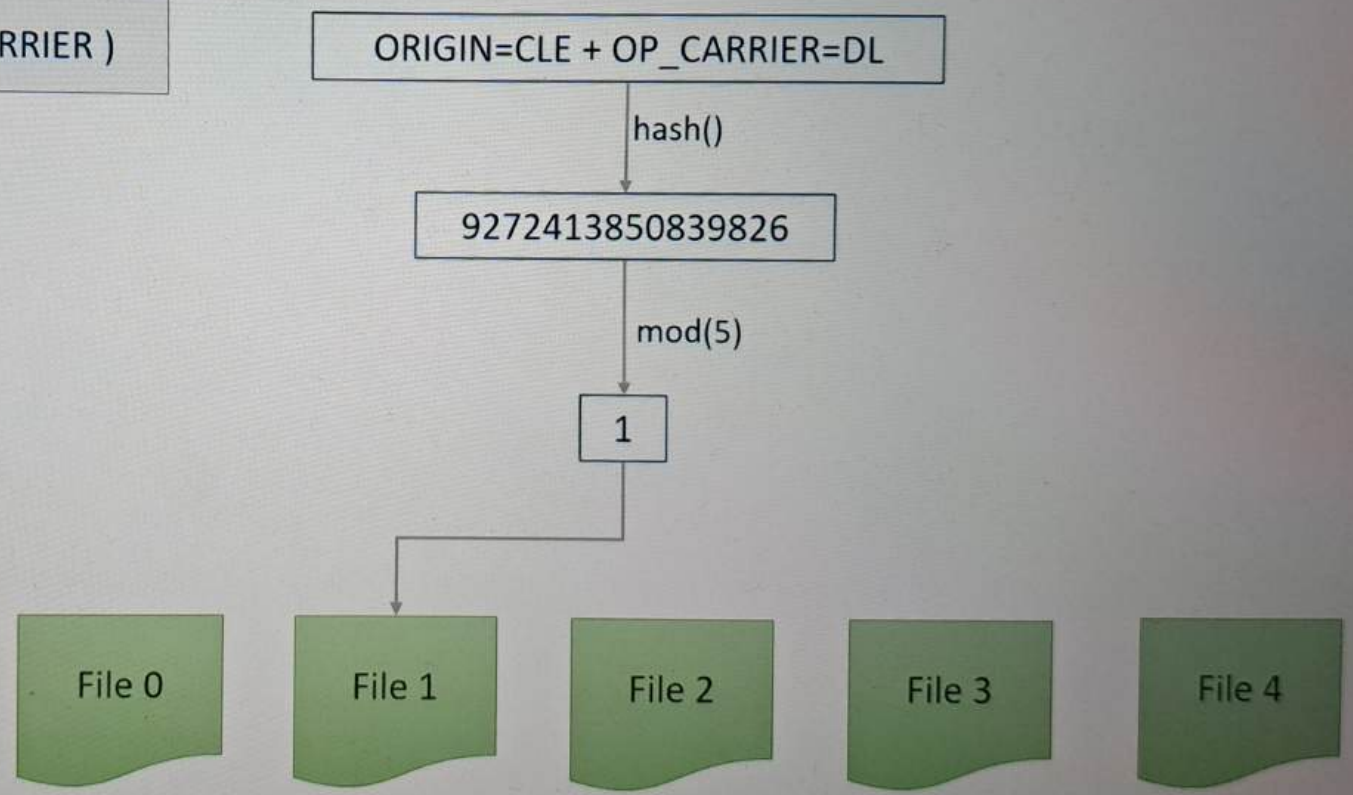
```
C:\Users\prashant\anaconda3\envs\pyspark\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/06-SparkSQLTableDemo/SparkSQLTableDemo.py
20/06/16 13:36:16 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
20/06/16 13:36:40 WARN HiveExternalCatalog: Persisting bucketed data source table `airline_db`.`flight_data_tbl` into Hive metastore in Spark SQL specific format, which is NOT com
20/06/16 13:36:41 INFO SparkSQLTableDemo: [[flight_data_tbl, airline_db, null, MANAGED, false]]

Process finished with exit code 0
```

9: Git | 6: TODO | 4: Run | Terminal | Python Console

Event Log

Spark Database and Tables

bucketBy(5, ORIGIN, OP_CARRIER)



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File Edit View Navigate Code Refactor Run Tools VCS Window Help 06-SparkSQLTableDemo - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00000.c000.csv - PyCharm

06-SparkSQLTableDemo > spark-warehouse > airline_db.db > flight_data_tbl > part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d...

Project

- 06-SparkSQLTableDemo C:\Users\prashant\Documents\GitHub\Spark-Programming-In-Python\06-Spark
  - app-logs
  - dataSource
  - lib
  - metastore_db
  - spark-warehouse
    - airline_db.db
      - flight_data_tbl
        - _SUCCESS.crc
        - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00000.c000.csv.crc
        - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00001.c000.csv.crc
        - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00002.c000.csv.crc
        - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00003.c000.csv.crc
        - .part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00004.c000.csv.crc
        - _SUCCESS
        - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00000.c000.csv
        - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00001.c000.csv
        - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00002.c000.csv
        - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00003.c000.csv
        - part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00004.c000.csv
  - derby.log
  - log4j.properties
  - SparkSQLTableDemo.py
- External Libraries
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SparkSQLTableDemo.py | part-00000-807e7005-ae0a-48f2-8184-3f885570fc7d_00000.c000.csv

The file size (6.29 MB) exceeds configured limit (2.56 MB). Code insight features are not available.

DL+CL 18/307

```
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2000-01-03,DL,1133,DCA,"Washington, DC",ATL,"Atlanta, GA",700,657,859,11,859,910,0,547
```

9: Git 6: TODO 4: Run Terminal Python Console Event Log

Navigate to the next occurrence 15 chars 3478:27 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

```
flightTimeParquetDF = spark.read \
    .format("parquet") \
    .load("dataSource/")

spark.sql("CREATE DATABASE IF NOT EXISTS AIRLINE_DB")
spark.catalog.setCurrentDatabase("AIRLINE_DB")

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    .format("csv") \
    .mode("overwrite") \
    .bucketBy(5, "OP_CARRIER", "ORIGIN") \
    .sortBy( "OP_CARRIER", "ORIGIN") \
    .saveAsTable("flight_data_tbl")

logger.info(spark.catalog.listTables("AIRLINE_DB"))
```